The purpose of this lecture is to understand the definition of Laplace transform.

$\frac{d}{dx}$, $\int (\,)\,dx$ etc. are one kind of operator that operates on a function and produces a new function.

original

$$\frac{d}{dx}(x^2) = 2x$$ ← new function

$$\int x^2\,dx = \frac{x^3}{3} + C$$ ← new function

original

These operators are linear, such as

$$\frac{d}{dx}\left[\alpha f(x) + \beta g(x)\right] = \alpha f'(x) + \beta g'(x)$$

$$\int \left[\alpha f(x) + \beta g(x)\right] dx = \alpha \int f(x)\, dx + \beta \int g(x)\, dx$$

That is these operators can be distributed linearly on linear combinations of functions.

Laplace transform operator has similar linearity property.

Laplace transform is a special type of integral transform.

Let's consider the below definite integral.

$$\int_1^2 \underbrace{2xy^2}_{f(x,y)}\,dx = \int_1^2 2y^2 \cdot x\,dx$$

$$= \int_1^2 f(y) \cdot x\,dx$$

$$= f(y) \int_1^2 x\,dx$$

$$= f(y) \left\{ \frac{x^2}{2} \Big|_1^2 \right\}$$

$$= f(y) \cdot \frac{3}{2}$$

$= f_1(y) =$ only function of y

Note that with this definite integral $f(x,y)$ got converted into

$f(y)$.

Laplace transform similarly converts a time domain function into a frequency domain function.

$$\int_a^b K(s,t) f(t)\, dt$$ transform f of variable t into F of variable s.

$$\int_0^\infty K(s,t) f(t)\, dt = \lim_{b \to \infty} \int_0^b K(s,t) f(t)\, dt \quad \text{— (1)}$$

If the limit in (1) exists, then we say that the integral exists or convergent. If the limit does not exist, the integral does not exist and is divergent.

function $K(s,t)$ is called the kernel of the transform.

when $K(s,t) = e^{-st}$ the integral is called Laplace transform.

Let f be a function defined for $t \geq 0$.

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$ is said to be the Laplace transform, provided the integral exists.

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st} \cdot 1 \, dt$$

$$= \lim_{b \to \infty} \left. \frac{e^{-st}}{-s} \right|_0^b$$

$$= \lim_{b \to \infty} \left[-\frac{e^{-sb}}{s} + \frac{e^0}{s} \right]$$

$$= \lim_{b \to \infty} -\frac{1}{e^{sb} \cdot s} + \frac{1}{s}$$

$$= 0 + 1/s = 1/s$$

$$\mathcal{L}\{1\} = \frac{1}{s}$$